

My whiteboard writing from week 4

Lecture 4 is thoroughly discussed in the notes and I abstain from reproducing most of my writing. However I will discuss in more details the calculations related to the confidence ellipsoid.

a) Case of Σ unknown $\rightarrow$ needs to be estimated

If using $S = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})'$ then $T^2 = n(\bar{X} - \mu_0)' S^{-1} (\bar{X} - \mu_0)$

$\sim \frac{p(n-1)}{(n-p)} F_{p, n-p}$. An $(1-\alpha) 100\%$ confidence region

for μ is then defined by $R = \{ \mu / n(\bar{X} - \mu)' S^{-1} (\bar{X} - \mu) \leq \frac{p(n-1)}{n-p} F_{p, n-p}(\alpha) \}$

where $F_{p, n-p}^\alpha$ denotes the upper $\alpha, 100\%$ point of $f_{p, n-p}$

The region R as defined above, is an ellipsoid in $\mathbb{R}^p$.

The half-axes of this ellipsoid are along the eigenvectors of S^{-1} . If $S = \sum_{i=1}^p \lambda_i e_i e_i'$ is the spectral decomposition of S with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$ then $S^{-1} = \sum_{i=1}^p \frac{1}{\lambda_i} e_i e_i'$

In direction e_1 , for example, "how long can we go?" by starting from the center of the ellipsoid (which is at $\bar{X}$). If we "can go" on κ_1 distance this means that we "reach" with $\kappa_1 e_1$ the borderline surface, i.e.

$$\kappa_1 e_1' \cdot \sum_{i=1}^p \frac{1}{\lambda_i} e_i e_i' \cdot \kappa_1 e_1 = \frac{p(n-1)}{n(n-p)} F_{p, n-p}(\alpha) \text{ must hold.}$$

Because $e_i' e_i = 1$ and $e_i' e_j = 0, j=2, 3, \dots, p$ we get:

$$\kappa_1^2 \cdot \frac{1}{\lambda_1} = \frac{p(n-1)}{n(n-p)} F_{p, n-p}(\alpha), \text{ i.e.}$$

$$\kappa_1 = \sqrt{\lambda_1} \sqrt{\frac{p(n-1)}{n(n-p)} F_{p, n-p}(\alpha)}$$

describes the length of the half-axis

Similarly, along each axis determined by the eigenvectors $e_i, i=1, 2, \dots, p$, we can go from the center $\bar{X}$ on a distance $K_i = \sqrt{\lambda_i} \sqrt{\frac{p(n-1)}{n(n-p)} F_{p, n-p}(\alpha)}$ in both

directions (e_i and $-e_i$). The values K_i describe the lengths of the half-axes.

b) Case of Σ known. $\rightarrow$ then we do not need to estimate it. Now the region would be $\tilde{K} = \{ \mu \mid n(\bar{X} - \mu)' \Sigma^{-1} (\bar{X} - \mu) \leq \chi^2_{p, \alpha} \}$

Similarly to a), if $\sum_{i=1}^p \lambda_i e_i e_i'$ is the spectral decomposition of Σ , by repeating the same steps like in a) we will end up with a confidence ellipsoid with a center again at $\bar{X}$ and with half axis-length $K_i = \sqrt{\lambda_i} \sqrt{\chi^2_{p, \alpha}}$ where $\chi^2_{p, \alpha}$ is the upper $\alpha \cdot 100\%$ point of the χ^2_p distribution.

In the attached extract from JW, illustration of the above calculations for $p=2$ in case a) is done:

For $n=42, \bar{X} = \begin{pmatrix} .564 \\ .603 \end{pmatrix}, S = \begin{pmatrix} .0144 & .0117 \\ .0117 & .0146 \end{pmatrix}$ we get $\lambda_1 = .026$
 $\lambda_2 = .002$

and $e_1 = \begin{pmatrix} .704 \\ .710 \end{pmatrix} \quad e_2 = \begin{pmatrix} -.710 \\ .704 \end{pmatrix}$. From tables, if $\alpha = 0.05$,

$F_{2, 40}(0.05) = 3.23$. The lengths of the two half-axes

are then $\sqrt{\lambda_1} \sqrt{\frac{p(n-1)}{n(n-p)} F_{p, n-p}(\alpha)} = \sqrt{.026} \sqrt{\frac{2(41)}{42(40)} 3.23} = .064$

and $\sqrt{\lambda_2} \sqrt{\frac{p(n-1)}{n(n-p)} F_{p, n-p}(\alpha)} = \sqrt{.002} \sqrt{\frac{2(41)}{42(40)} 3.23} = .018$

The axes are along e_1 and e_2 going out from the center of the ellipse at $\begin{pmatrix} .564 \\ .603 \end{pmatrix}$.

